SZABIST
University

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ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

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Total Marks:________

Obtained Marks: ________

Computing Vision

Lab Task # 02

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Windows Python Environment Install and Setup

Step 1: Download and Install Python

Installing Anaconda Navigator

Create a Jupyter Notebook File (.ipynb)

Basic Operation on Jupyter Notebook

Solution:

Windows Python Environment Setup (Complete Beginner Guide)

This repository provides a detailed walkthrough for setting up a complete Python development environment on Windows. It includes installation of Python, Anaconda, and creation of a Jupyter Notebook file for interactive coding.

Overview of Setup Process

The setup follows this sequence:

1. Install Python
2. Configure System Path
3. Install Anaconda Navigator
4. Launch Jupyter Notebook
5. Create and Run a Python Notebook

Step 1: Install Python

◇ **Download Python**

- Visit: https://www.python.org/downloads/
- Download the latest stable version.

◇ **Installation Steps**

- Run the installer.
- ✓ Check Install for all users
- ✓ Check Add Python to PATH
- Click Install Now
- When prompted, click Yes to allow installation.
- Disable the Path Length Limit to avoid path errors.

◇ Why This Is Important

Adding Python to PATH allows you to run Python from:

- Command Prompt
- PowerShell
- Any IDE or Notebook environment

Step 2: Install Anaconda Navigator

Anaconda simplifies Python package management and includes essential tools like Jupyter Notebook.

◇ Download Anaconda

- Visit: https://www.anaconda.com/download
- Click Download
- Skip registration if prompted

◇ Installation Configuration

- Click Next
- Accept License Agreement
- Select Just Me installation (recommended)
- Choose installation directory (avoid spaces and special characters)
- Choose whether to register Anaconda as default Python
- Click Install

After installation, click Finish.

Step 3: Launch Jupyter Notebook

1. Open Anaconda Navigator
2. Locate Jupyter Notebook
3. Click Launch
4. Your browser will open a local server interface
5. Select your working directory
6. Click New → Python 3
7. A new .ipynb file will open

You can now:

- Write Python code
- Run code cells

- Add text explanations
- Save your notebook

Final Result

After completing this setup, you will have:

✓ Python installed and configured

✓ Environment variables properly set

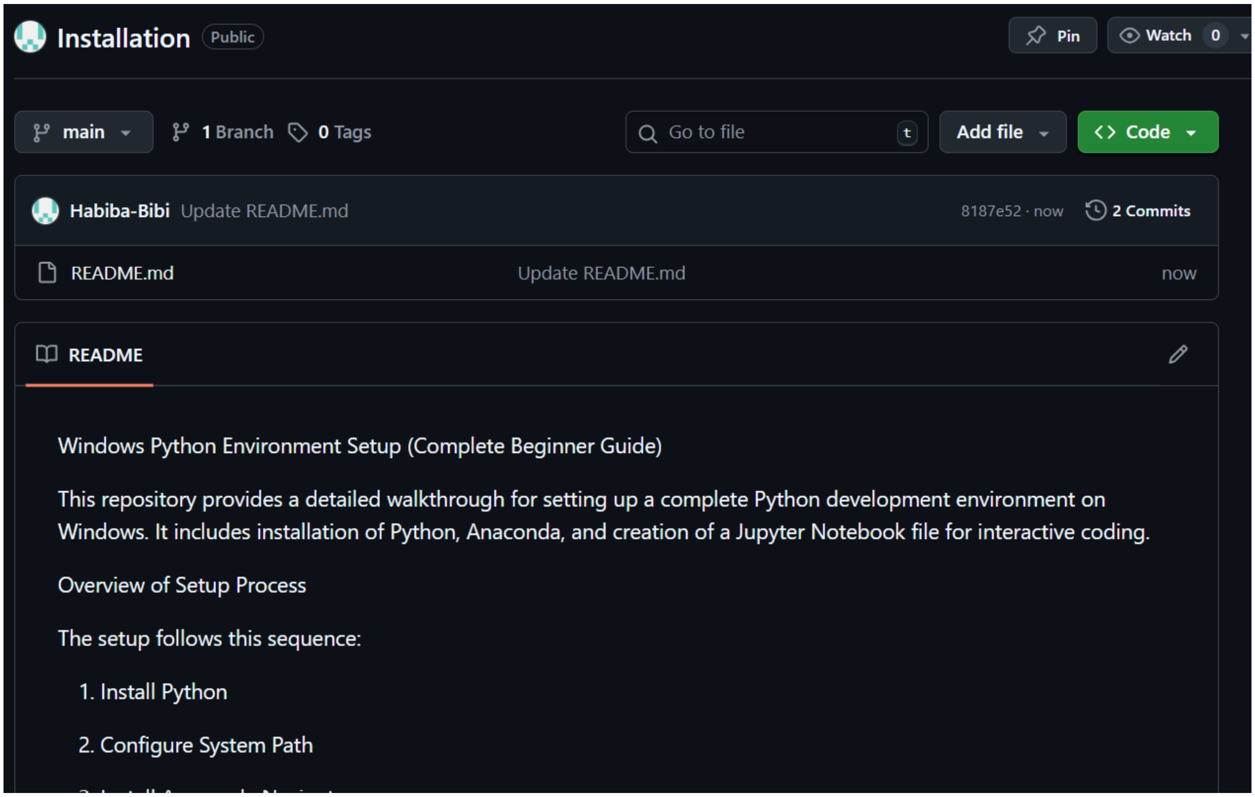
✓ Anaconda package manager ready

✓ Jupyter Notebook for interactive coding

✓ A fully functional development environment

Recommended For

- Computer Science Students
- Data Science Beginners
- Machine Learning Labs
- Academic Projects
- Python Programming Starters



https://github.com/Habiba-Bibi/